

Lab - Document Enterprise Cybersecurity Issues

Objectives

Part 1: Record your assessment of Athena's cybersecurity issues.

Part 2: Record the different types of assets owned by Athena.

Part 3: List the threats for each asset type.

Part 4: Recommend mitigation techniques to address each threat.

Note: This lab assumes you have basic knowledge of vulnerabilities, threats, and mitigation techniques.

Scenario

Athena Learning Incorporated is an educational service provider. Athena has two major lines of business: course content creation and online learning services. Athena creates learning content and hosts learning content. Athena also provides internet sales services that enable its partners to charge their students to attend their courses.

Athena employs about 100 people in its headquarters office, and about 5 people each in its London and Singapore offices. Because it provides content and delivery services globally, Athena must comply with diverse privacy and security standards.

Athena serves as custodian for its own content and content that belongs to its partners. That content includes text, graphic, video, and interactive assets. This content is the essential intellectual property of the company. It also manages student account information including student registration, authentication, records, and payment information. Athena manages its own SQL databases, some of which are connected to web portals.

The Athena network consists of mostly MS Windows and Apple IOS clients with a mix of Microsoft and Linux servers to store business and employee records, learning content assets, and financial information, including customer data. The hosts include various PC brands and models of varying age. Different versions of operating systems are in use. Athena uses cloud services to deliver courses to the public, but must house assembled courses on the internal network for creation and editing. When the courses become available, they are mirrored to the cloud. Employees are permitted to use their personal phones and tablets for work. In addition, some employees work from home, but require full network access to do so. Athena also hosts its own DNS, email, and intranet services.

Athena employees use common office application software, custom applications, and tools that have been created internally.

Athena provides access to parts of its internal network to its partners through a secure web portal. Clients are able to preview their course content and deliver course assets to Athena for assembly in the Athena learning management system. Students interact with the cloud-managed learning platform through their web account logins.

In this lab, you will apply your knowledge of cybersecurity threats and mitigation techniques to a corporate setting. You will read about a business, classify its assets, and then list the potential vulnerabilities and threats

that the business faces. Finally, you will recommend threat mitigation measures for the threats that you identify.

Required Resources

= Devices with internet access

Instructions

Part 1: Record your assessment of Athena's cybersecurity issues.

Study the **Scenario** above about Athena Learning Incorporated. Focus on identifying the data, software, hardware, and network assets that need to be protected to ensure that company is not impacted by various types of threats that have been discussed in the course so far.

Use the tables below to record your answers.

Information/Data Assets	Threats	Mitigation
Servers that store personal data belonging to students	Data theft of protected information, Identity theft	Ecrypt store data
Data and service requirements for staff working from home	Data theft, Data tampering	VPN
Educational resources available through the online course site	Data theft, Unauthorized access, Alteration of content	Hardening of web application, Strong Authentication
Sales data related to students	Theft of credit card information, Identity theft	Ecrypt store data
Customer assets in the process of being uploaded to Apollo	Data theft, Data tampering	

Software Assets	Threats	Mitigation
Office Applications	Various application attacks, Exploits of unpatched vulnerabilities	Security policies regarding use of unauthorized software, Patching
Web Server Software	Cross site scripting, Vulnerable server software	Patching
SQL Databases	Data entry errors, XML or SQL injections	Data input validation
Host Operating Systems	Security Vulnerabilities, Phishing, Malware, Malicious websites	Patching, Antivirus software, Block access to known malicious sites, Use security training

Software Assets	Threats	Mitigation
In House Applications	Various application exploits	Strong input validation, Secure software development

Physical Assets	Threats	Mitigation
Networking Equipment	Physical damage, Unauthorized administrative access, Power interruption	Physically secure wiring closets and equipment locations, Power protection
Laptop PCs	Lost, theft, or damaged, Hard drive failure, Other physical damage	Antivirus, Drive encryption, Physical access control, Host based firewall, Regular data backups
Desktop PCs	Other physical damage, Control access to facilities, Hard drive failure, Power interruption	User security training, Data backups, Power protection, Badge based or biometric access control
File Servers	Hard drive failure, Physical damage, Power interruption	Power Protection, Physically secure server room, redundant servers, Automated Back ups

Network Assets	Threats	Mitigation
Connection to ISP	DoS, DDoS	Block external ICMP packets with firewalls
Wired LAN	Unauthorized access, Man in the Middle, Detect unauthorized network scanning	Strong passwords
Wireless LAN	Rogue access points, Unauthorized access, Man in the Middle	VPN for remote wireless users, Access point placement, Use tools to detect rogue access points, Use strong authentication and encryption
IP Services	DNS spoofing address	Next generation firewalls

Part 2: Record the different types of assets owned by Athena.

From the information in the **Scenario**, and your knowledge of business in general, fill in the first columns of each table with the relevant assets that are owned by Athena. You should have at least three entries in each table.

The different types of assets are defined as follows:

- = **Information/Data Assets** - any data that is used by the company, in any of the three states of data. This data could be Athena's business data, Athena's learning content, student sales and learning data, or partner data.
- = **Software Assets** - any software that is used by Athena, including commercial business applications, operating systems, server software, database software, and custom software.
- = **Physical Assets** - the physical devices, equipment, and other property that are used by Athena in the course of their business.
- = **Network Assets** - the types of networks and network connections that are hosted or used by Athena in the course of its business.

Part 3: List the threats for each asset type.

- a. Review the information that you have learned in this pathway regarding vulnerabilities and threats.

Question:

What is the difference between a threat and a vulnerability? A threat is any event, actor, or condition with the potential to cause harm by exploiting a weakness in a system, process, or environment. A vulnerability is a flaw, weakness, or gap in a system, process, or control that could be exploited by a threat to cause harm.

- b. Complete the second column of the table with threats that could exploit vulnerabilities for each asset that you listed. There is usually more than one threat to each asset.

Part 4: Recommend mitigation techniques to address each threat.

Review the information that you have learned so far about ways to mitigate various cybersecurity threats. Complete the third column of the table with mitigation techniques that can be done to avoid or limit the damage caused by each potential threat.

Reflection

1. Why is it useful to categorize assets when identifying threats and mitigation techniques?

Classifying assets by type provides a structured foundation for identifying relevant threats. Without this categorization, the sheer volume and diversity of assets can make it difficult to begin a meaningful threat analysis.

2. Do some threats have the same or similar mitigation measures? Why is it important to note this? Yes, some threats can be mitigated using the same means. For example, a system of updating host and server software can help to mitigate threats to customer and company data. VPNs can encrypt data uploaded to Athena's servers by customers and can also protect assets sent to the cloud. Knowing this helps guide the choice and implementation of threat mitigation solutions.
3. What have you learned about the application of knowledge of cybersecurity threats and mitigation techniques to the context of a simulated organization? A comprehensive cybersecurity program relies on multiple, layered security measures that work together to protect an organization's diverse assets. Because no single control

can address the full spectrum of modern threats, effective protection requires a combination of complementary safeguards operating across people, processes, and technology.